

Data Description Document

Introduction

This dataset pairs real and simulated production data obtained from an aerospace surface treatment facility. Its main objective is to furnish researchers with a reproducible benchmark for studies in production scheduling, capacity planning, and optimization. By combining authentic shop-floor records with synthetically aggregated scenarios, the dataset supports the replication of computational experiments and the development of new scheduling models.

The real data were extracted directly from the plant's manufacturing execution system (MES), capturing daily demand patterns exactly as they occurred. Simulated scenarios were generated by summing demand from two consecutive production days, thereby introducing higher variability and stress-testing scheduling algorithms. These dual datasets were used in computational experiments to evaluate a Mixed-Integer Linear Programming (MILP) scheduling model. The related research article analyzes the model's performance under both real and aggregated demand conditions.

Dataset Overview

This dataset was organized to support capacity planning and production scheduling studies for surface treatment in the aerospace industry. It contains information about:

1. Maximum quantity of an item in a fixture;
2. Maximum number of fixtures per item in each tank;
3. Raw material of each item;
4. Daily real demand of each item (30 separate files);
5. Daily simulated demand of each item (30 separate files).

Each '.csv' file can be linked through the common 'item_ID' field.

Maximum Quantity per Fixture

This file records, for every distinct item_ID, the maximum number of units that can be mounted on a single fixture used in the surface-treatment line. The limit is

determined by geometric constraints and fixture design, and it is a fixed parameter in all scheduling and capacity-planning models.

File: MaxPerFixture.csv

Attribute	Type	Description
item_ID	String	Item identification code (e.g., "830D0100")
max_per_fixture	Integer	Maximum number of item units that fit in a fixture

Maximum Number of Fixtures per Item in Each Tank

This file links every item_ID to the maximum number of fixtures that can physically fit into each tank during a single loading cycle. The limit varies with the overall dimensions and geometry of the part-fixture assembly, which determine how many fixtures can be arranged inside a given tank footprint.

File: MaxFixturesPerTank.csv

Attribute	Type	Description
item_ID	String	Item code
tank_B1	Integer	Maximum number of fixtures allocated to the item in that tank
tank_B2	Integer	Maximum number of fixtures allocated to the item in that tank

Raw Material of Each Item

Although classical relational design would store raw material in a normalized table such as Item_ID $\rightarrow$ RawMaterial, we intentionally encoded it using one-hot binary columns (B1, B2, B6, LTS30) for three practical reasons:

1. Direct integration with optimization models: In our MILP scheduling experiments these same 0/1 indicators appear as model parameters. Keeping the data in one-

- hot form eliminates preprocessing steps and prevents mismatches between the database and the solver inputs.
2. Faster ingestion by analytics tools: Numerical columns are parsed more efficiently than categorical strings by common analytics packages (pandas, R, BI platforms). With a small, mutually exclusive domain, each item has exactly one raw material, filters, joins, and aggregations become simpler and less costly.
 3. Single raw-material assignment per item: Because every item belongs to exactly one raw-material class, the one-hot layout introduces no logical redundancy: each row contains a single “1.” This guarantees data integrity without extra foreign keys or validation while preserving a flat-file structure compatible with repository requirements.

File: RawMaterial.csv

Attribute	Type	Description
item_ID	String	Item code
B1	Integer	1 = item is made from raw material B1; 0 = otherwise
B2	Integer	1 = item is made from raw material B2; 0 = otherwise
B6	Integer	1 = item is made from raw material B6; 0 = otherwise
LTS30	Integer	1 = item is made from raw material LTS30; 0 = otherwise

Real Daily Demands

This folder contains the actual daily demand captured directly from the plant’s Manufacturing Execution System (MES) during the July-to-August data-collection window. Each file represents one production day and lists the exact quantities ordered for every item_ID.

Folder: Real_Demand

Files: 07_10.csv, 07_11.csv, ..., 08_20.csv

Attribute	Type	Description
item_ID	String	Item code
quantity	Integer	Quantity demanded on that day (real demand)

Simulated Daily Demands

To test the robustness of scheduling algorithms under higher variability, we generated 30 synthetic demand scenarios by aggregating demand from two consecutive real days and randomly permuting item order. Each file contains the simulated quantities for a single scenario while retaining the same column structure as the real-demand files

Folder: Simulated_Demand

Files: 1.csv, 2.csv, ..., 30.csv

Attribute	Type	Description
item_ID	String	Item code
quantity	Integer	Quantity demanded on that day (real demand)

> Note: file names follow the convention XX.csv, where XX ranges from 01 to 30.

File Relationships

Figure 1 depicts how every file and folder in the dataset is linked through the common identifier Item_ID. Acting as a primary key, this field ties the static parameter tables, MaxPerFixture.csv, MaxFixturesPerTank.csv, and RawMaterial.csv, to the time-dependent demand records stored in the Real_Demand and Simulated_Demand folders. With Item_ID as the joining field, capacity analyses and production-scheduling experiments can seamlessly merge fixture limits, tank constraints, raw-material attributes, and daily or scenario-based demand profiles into a unified analytical view.

